OPERATING SYSTEMS
UE24CS242B

Mutual Exclusion and Synchronization

Pavan A C
Department of Computer Science & Engineering

Slides Credits for all the PPTs of this course

- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating

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Software Approaches

Pavan A C
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Background

- Processes can execute concurrently
  - May be interrupted at any time, partially completing execution
- Concurrent access to shared data may result in data inconsistency
- Maintaining data consistency requires mechanisms to ensure the orderly execution of cooperating processes
- Illustration of the problem:
  Suppose that we wanted to provide a solution to the consumer-producer problem that fills ***all*** the buffers. We can do so by having an integer **counter** that keeps track of the number of

Producer

```
while (true) {
        /* produce an item in next_produced */


        while (counter == BUFFER_SIZE) ;
                /* do nothing */
        buffer[in] = next_produced;
        in = (in + 1) % BUFFER_SIZE; /* Variable in points to the
next free position in the buffer */
        counter++;
}
```

Consumer


```
while (true) {
        while (counter == 0)
                ; /* do nothing */
        next_consumed = buffer[out];
        out = (out + 1) % BUFFER_SIZE; /* Variable out points to the
first full position in the buffer */
                counter--;
        /* consume the item in next consumed */
}
```

Race Condition

- **counter++** could be implemented as

```
register1 = counter
register1 = register1 + 1
counter = register1
```

- **counter--** could be implemented as

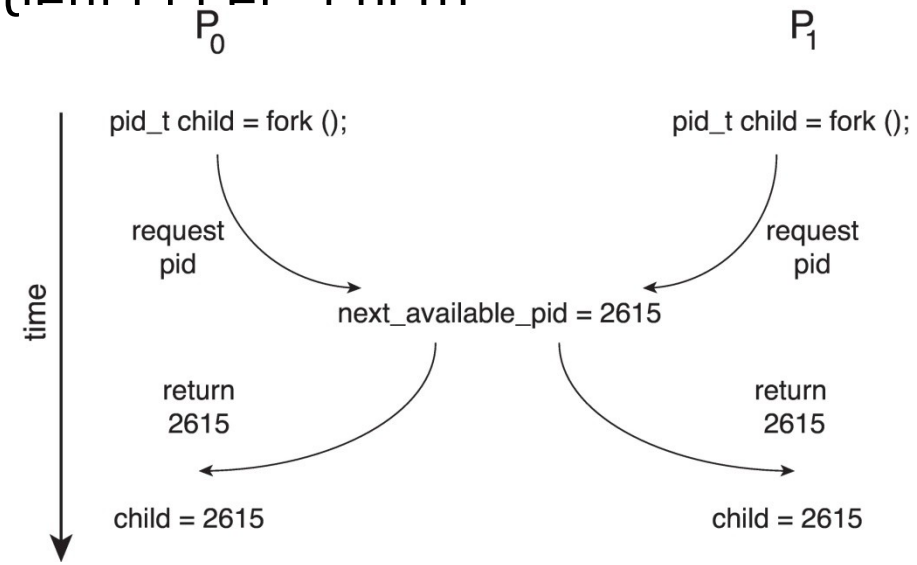
```
register2 = counter
register2 = register2 - 1
counter = register2
```

- Consider this execution interleaving with "count = 5" initially:

S0: producer execute **register1 = counter** {register1 = 5}
S1: producer execute **register1 = register1 + 1** {register1 = 6}
S2: consumer execute **register2 = counter** {register2 = 5}

Race Condition (Another Example)

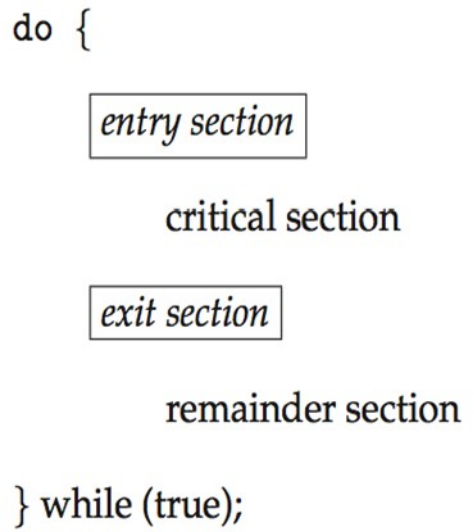
- Processes P_0 and P_1 are creating child processes using the fork() system call
- Race condition on kernel variable next_available_pid which represents the next available process identifier (pid)



- Unless there is a mechanism to prevent P_0 and P_1 from

Critical Section Problem

- Consider system of ***n*** processes $\{p_0, p_1, \ldots p_{n-1}\}$
- Each process has **critical section** segment of code
  - Process may be changing common variables, updating table, writing file, etc
  - When one process in critical section, no other may be in its critical section
- ***Critical section problem*** is to design protocol to solve this
- Each process must ask permission



Critical Section

- General structure of process P_i

```
do {
        entry section
            critical section
        exit section
            remainder section
} while (true);
```

Solution to Critical-Section Problem

1. **Mutual Exclusion** - If process P_i is executing in its critical section, then no other processes can be executing in their critical sections
2. **Progress** - If no process is executing in its critical section and there exist some processes that wish to enter their critical section, then the selection of the processes that will enter the critical section next cannot be postponed indefinitely
3. **Bounded Waiting** - A bound must exist on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its

Critical-Section Handling in OS

Two approaches depending on if kernel is preemptive or non-preemptive

- **Preemptive** – allows preemption of process when running in kernel mode.
  - Preemptive kernels are difficult to design on SMP architectures
- **Non-preemptive** – runs until exits kernel mode, blocks, or voluntarily yields CPU
  - Essentially free of race conditions in kernel mode

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Peterson's Solution

- Software-based solution to the Critical Section problem
- Good algorithmic description of solving the problem
- Assume that the **load** and **store** machine-language instructions are atomic; that is, cannot be interrupted
- **Peterson's Solution** restricted to two process.
- P_i and P_j are two process, j=1-i
- Peterson solution requires two processes share two data items:
  - int turn;
  - Boolean flag[2]
- The variable **turn** indicates whose turn it is to enter the critical section
- The **flag** array is used to indicate if a process is ready to enter the critical

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Algorithm for Process P_i

```
while (true) {
        flag[i] = true;
        turn = j;
        while (flag[j] && turn = = j);
                /* critical section */
        flag[i] = false;
                /* remainder section */
}
```

The structure of process P_i in Peterson's solution

Structure of process P_i in Peterson's solution

```
do {
    flag [i] = true ;
    turn = j ;
    while ( flag [ j ] && turn == [ j ] ) ;

        critical section

    flag [i] = false ;

        remainder section

} while (TRUE) ;
```

Structure of process P_j in Peterson's solution

```
do {
    flag [ j ] = true ;
    turn = i ;
    while ( flag [ i ] && turn == [ i ] ) ;

        critical section

    flag [ j ] = false ;

        remainder section

} while (TRUE) ;
```

Software Solution

- To prove solution is correct, we need show
- Mutual exclusion is preserved

  P_i enters critical section only if:

  turn = i

  and **turn** cannot be both 0 and 1 at the same time
- What about the Progress requirement?
- What about the Bounded-waiting requirement?

Correctness of Peterson's Solution

- Provable that the three CS requirement are met:

  1. Mutual exclusion is preserved

     P_i enters CS only if:

     either **flag[j] = false** or **turn = i**

  2. Progress requirement is satisfied

  3. Bounded-waiting requirement is met

THANK YOU

Pavan A C
Department of Computer Science & Engineering

pavanac@pes.edu